

ANVAY VATS

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EDUCATION

University of Waterloo (Triple Major)

Sep 2022 – Apr 2027

Bachelor of Computer Science, Combinatorics and Optimization, and Pure Mathematics

- Clubs: Wat.AI, Wat Street (Quant Finance), CS Club, Data Science Club, Pure Math Club, Statistics Club
- CS Courses: DS & Algorithms, Operating Systems, Assemblers (RISC-V), Neural Networks, Applied Cryptography
- Math Courses (**all advanced**): Calculus I, II, & III; Linear Algebra I & II; Probability, Statistics, Groups, Optimization

TECHNICAL SKILLS

Languages: Python, Rust, C++20, Go, SQL, TypeScript, Julia, Elixir

AI/ML: PyTorch, Tensorflow, JAX, HuggingFace, LangGraph, CrewAI, DSPy, MLFlow, CUDA, Weaviate

Data/Systems: GraphQL, DuckDB, Redis, Kafka, NATS, QuantLib, Kafka, Risk Models, Timeseries Analysis

DevOps/Cloud: Docker, Terraform, Jenkins, Airflow, AWS (Lambda, EC2, ECS, SageMaker), Kubernetes

Interests: **Blogging**, Startups (0-1 builds), Philosophy, Mental math, Table Tennis, Poker, Gaming, Stand-up

EXPERIENCE

Founding AI Engineer | **Cupertino Labs (Vercel AI Accelerator S25)**

Apr 2025 – Aug 2025

CrewAI, LangGraph, MCP, Manus, tRPC, Jenkins, SonarQube, SageMaker, Airflow

San Francisco

- Shipped the analytics engine used on **10 million+ smartphones worldwide**, in contracts with AT&T and Airtel
- Replaced ETL on GCP clusters with a **CrewAI + Manus multi-agent workflow**, cutting KPI reporting latency by **90%**
- Built **DevSecOps Jenkins pipelines** integrating SonarQube (coverage, SCA, quality gates), Fortify, and JFrog Xray

Software Developer | **DataCurve (YC W24)**

Oct 2024 – Dec 2024

Elixir, Ecto, GraphQL, TypeScript, NextJS, TablePlus, Postman, Clerk

San Francisco

- Managed a **7-figure contract** to supply clean training data for **Cohere's Chain-of-Thought LLM evaluation**
- Applied Ecto Repos, queries, and schemas with Elixir resolvers and schemas to ingest **1000+ user submissions**
- Added features for users to do real-time voice recordings to more accurately emulate human thought processes

Lead Developer (Gen AI) | **Arvo AI (Panache VC-backed)**

Aug 2024 – Dec 2024

LangGraph, React, Ollama, llama.cpp, MLFlow, Weaviate, Docker, Terraform

Montreal

- Led design + development of a freight logistics scheduler on a **6-figure contract** using LangGraph and Ollama
- Made **PoWs, DoDs, GANTT, RACI, and RAID logs** and finished 150+ tickets on Linear in Agile Kanban sprints
- Delivered GNN-based freight optimizer and RAG pipeline with **90% precision, 95% recall, and 100 ms latency**

Quantitative Developer | **TIW Capital (\$200M+ AUM)**

Apr 2024 – Aug 2024

Python, C++, Julia, Redis, Kafka, QuantLib, PostgreSQL, GoRoutines

Singapore

- Implemented **dynamic risk optimization** systems using **variable position sizing** and cross-sectional momentum strategies, reducing drawdowns by 32% while maintaining alpha
- Engineered high-frequency **adaptive trend following** algorithms with multi-asset **normalized signal processing**
- Developed **carry-enhanced, fast mean-reversion strategies** for break-out detection based in **deep diversification**

Machine Learning Researcher | **iLoF**

Jan 2024 – Apr 2024

Python, FastAPI, Redis, Celery, Docker, Kubernetes, PostgreSQL, AWS Lambda

London

- Made synthetic data engine using VAEs, copulas, and probabilistic PCA, **raising cancer detection recall by 19%**
- Created domain adaptation pipeline with generative models on K8, increasing cross-cohort accuracy by 28%
- Deployed async workers for Bayesian ensemble, cutting photonics false positives by 23% to **scale 10M+ signals**

AWARDS

Silver Medal – National Olympiad of Informatics (NOI), Singapore

Feb 2022

\$25000 Global Math Scholarship

Sep 2022

PROJECTS

Dynamis Trading System (\$3.7M USD Monthly Volume Traded)

Rust, Python, Redis, Polars, JAX, AWS (ECS Fargate, EC2, Secrets Manager, TGW), NATS, CUDA

- Leading **full-stack development** in a team of 6 with the goal of **low-latency Solana trading strategies**
- Created the **fastest Twitter/X Solana token sniper in the world** with **25ms end-to-end purchase latency**
- Deployed **Langgraph agentic workflows** to transform plaintext trading strategies into **CFGs for parallel parsing**
- Architected data-intensive distributed system using **Jito gRPCs** and **NATS JetStreams** with microservices